For consumer API for freight booking, the following code lists are relevant:

- Document Reference Type Codes
- Contract Condition Codes
- Incoterm Descriptions
- Item Category Code
- Note Types
- Shipping Type Descriptions
- Time Zone
- Transport Means Description Code
- Transportation Modes
- Unit of Measure Date Time
- Unit of Measure Length
- Unit of Measure Quantity
- Unit of Measure Temperature
- Unit of Measure Volume
- Unit of Measure Weight

## Additional Information

- In the **Assign Codes** app, if you do not maintain the mapping between **Code in External System** and **LBN Code**, for the **TransportModeCode** attribute, when the ordering party sends a value from their external transportation system, the respective value that you receieve in SAP Business Network for Logistics might be different as the static mapping of values between these two systems are different.

Transport Mode Codes

| SAP Transportation Management System | SAP Business Network for Logistics System |
|---|---|
| 03 - Ocean | 01 - Ocean |
| 01 - Road | 03 - Road |
| 05 - Air | 04 - Air |

# Provider API for Freight Booking

As a carrier, you can use this inbound provider API to manage ocean or air freight booking request confirmation process when integrating SAP Business Network for Logistics with other systems.

## Description

You can use this inbound API to create a response for a freight booking in SAP Business Network for Logistics. With this functionality, you can confirm or reject a freight booking.

The ordering party sends a freight booking for the reservation of freight space from their transportation system to SAP Business Network for Logistics. If you have integrated the freight booking confirmation process using the **Consumer API for Freight Booking**, then it would be available in your backend system. You can respond to the freight booking by either confirming or rejecting using the **Provider API for Freight Booking**.

You can submit a payload with confirmation soon after you receive a freight booking. Later, you can update the details and resend your confirmation. For example, you can update the cut-off dates, VGM cut-off (for ocean freight booking only), arrival and departure dates for different stops, reference document numbers, and so on. Additionally, you can view the air freight bookings in **Confirm Freight Bookings (Air)** app and ocean freight bookings in **Confirm Freight Bookings (Ocean)** app in SAP Business Network for Logistics.

For more information, see Confirm Freight Bookings (Ocean) and Confirm Freight Bookings (Air).

## Connection Information

| Technical Name | Field |
|---|---|
| Company Type | Carrier |
| Version | 1.0.0 |
| Format | JSON |
| Dependencies | Your ordering party has sent the booking request from their Transportation Management on SAP S4/HANA system. |
| Base Document Type | Type of freight document supported:<br>• 1122 – Transportation Order |

You can download the YAML file from the **Provider API for Freight Booking** in SAP API Hub.

## Data Dictionary

In the SAP API Hub, in the **Provider API for Freight Booking**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
- Max length
- Required or not
- Example

## Relevant Codes

SAP Business Network for Logistics has defined its own code values for the various attributes used within business network. These codes are used during B2B (business-to-business) communication so using APIs in business network. For more details on the list of codes supported in SAP Business Network for Logistics, see Code Lists.

For consumer API for freight booking, the following code lists are relevant:

- Document Reference Type Codes
- Contract Condition Codes
- Incoterm Descriptions
- Item Category Code
- Note Types
- Shipping Type Descriptions
- Time Zone
- Transport Means Standard Code Descriptions
- Transportation Modes
- Unit of Measure Date Time
- Unit of Measure Length
- Unit of Measure Quantity
- Unit of Measure Temperature
- Unit of Measure Volume
- Unit of Measure Weight

## Possible Errors

- Incorrect master air waybill, which is outside of the defined stoke.
- If the airline code is not assigned (or incorrectly maintained) by shipper to the carrier, sending `JourneyID` (Flight Code is the UI field) in the provider API would fail at the shipper backend system in case of air freight booking.
- By sending `ItemIDs` that are not part of the consumer API will give an error in the provider API.
- The shipper can maintain the necessary settings to disable you from adding attachments. In this scenario, if you respond back to the shipper with an attachment, then the response fails.

# Freight Tracking

You can use the tracking process to keep your ordering parties informed about the transportation of their freight.

Using the freight tracking APIs, you can create or update tracking requests and post events.

You can perform the following activities using the APIs for freight booking:

- You can use the Provider API for Order Events (FC) to post events.
- You can use the Consumer API for Order Tracking (FC) to create, update, or delete a tracking request in an external system.

### More Information

Freight Orders for Reporting

# Consumer API for Order Tracking (FC)

As a carrier, you can use this consumer API to create or update a tracking request when integrating SAP Business Network for Logistics with other systems.

## Description

You can use this outbound API to perform the following:

- **Create Tracking Request**

  Enables you to create a tracking request in an external system.

- **Delete Tracking Request**

  Enables you to delete a tracking request in an external system.

# Provider API for Order Events (FC)

As a carrier, you can use this provider API to post events for an order. The events will be posted in SAP Business Network for Logistics and then forwarded to the system where the order was initially created.

## Description

You can use this inbound API to create an event in a tracked order in SAP Business Network for Logistics. You can use this API to post events to SAP Business Network for Logistics. With this, the shipper can see the events in their external system and in SAP Business Network for Logistics as well.

# Freight Settlement

Using the freight settlement API, you can perform freight settlement with your ordering party.

You can use the Provider API for Invoicing to create an invoice request.

## More Information

Manage Invoices

# Provider API for Invoicing

As a carrier, you can use this provider API to manage invoicing process when integrating SAP Business Network for Logistics with other systems.

## Description

You can use this inbound API to create an invoice.

The ordering party sends a freight order from their transportation system to SAP Business Network for Logistics. If you have integrated to the freight order confirmation process using the **Consumer API for Freight Subcontracting**, then it would be available in your backend system. You can create an invoice for this freight order using the **Provider API for Invoicing**.

You can submit a payload with the charges that match the freight order an invoice is created in SAP Business Network for Logistics in the carrier tenant. Additionally, you can view the freight order for invoicing in the **Invoice Freight Documents** app in SAP Business Network for Logistics. If there is any mismatch in the charges, the system creates a dispute either in the ordering party's SAP Business Network for Logistics ordering party tenant or the external transportation system depending on the settings that the ordering party has specified in their **General Settings** app.

### ℹ Note

For the ocean freight settlement process, by default, disputes are created in the shipper's SAP Transportation Management system irrespective of the settings specified in the **General Settings** app. However, you can manage the dispute process from your SAP Business Network for Logistics.

The system also creates an invoice corresponding to the freight order sent by the ordering party in the carrier tenant in SAP Business Network for Logistics.

You can view the freight order sent by the ordering party in the **Invoice Freight Documents** app and the corresponding invoice in the **Manage Invoices** app. For more information on how to manage the documents in these apps, see Invoice Freight Documents and Manage Invoices.

## Connection Information

| Technical Name | Field |
|---|---|
| Company Type | Carrier |
| Version | 1.0.0 |
| Format | JSON |
| Prerequisites | Ensure that your ordering party has maintained charge codes mapping in SAP Business Network for Logistics.<br><br>Provide the LBN ID of purchasing party and bill from party in payload. |
| Predecessors | Recommended: `LBNFreightOrderForSubcontracting` |
| Dependencies | Your ordering party has sent the freight document to be invoiced by you to SAP LBN. |
| Base Document Type | Type of freight document supported:<br>• 1122 – Transportation Order<br>• 610 – Transportation Booking Order |

You can download the YAML file from the Provider API for Invoicing in SAP API Hub.

## Data Dictionary

In the SAP API Hub, in the **Provider API for Invoicing**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
- Max length
- Required or not
- Example

## Relevant Codes

SAP Business Network for Logistics has defined its own code values for the various attributes used within business network. These codes are used during B2B communication so using APIs in business network. For more details on the list of codes supported in SAP Business Network for Logistics, see Code Lists.

For invoice submission, the following code lists are relevant:

- **Charge Code Descriptions**
- **Document Reference Type Codes**(For invoicing, codes1122 and 610 are supported)
- **Unit of Measure Date Time**
- **Unit of Measure Length**
- **Unit of Measure Quantity**
- **Unit of Measure Temperature**
- **Unit of Measure Volume**
- **Unit of Measure Weight**
- **Currency codes**

## Additional Information

- For supported date format and examples, download the YAML file.
- Post submission of the API message, login to SAP Business Network for Logistics and launch the **Manage Integration Logs** app. If you see a message with status code 200, then the submission may not be successful because there might be data errors or business validation errors that prevented the creation of invoice in SAP Business Network for Logistics.
- Either the `freightDocumentId` (`freightOrderId` or `freightBookingId`) or the `alternateIDCategory` is a mandatory field.
- `alternateIDCategory` – If you enter a value for this field, then this value is used to identify the freight document. Note that if the `freightDocumentId` field is also filled, then this value takes precedence.

  The following table contains the allowed values:

| Alternate ID Category | Description |
| --- | --- |
| MBL_CONTAINERID | When you use this category, you must provide the `alternateID` with the following values:<br>◦ `typeCode` as T52 (Master Bill of Lading)<br>◦ `typeCode` as CONT (Container ID) |
| BOOKINGREF_CONTAINERID | When you use this category, you must provide the `alternateID` with the following values:<br>◦ `typeCode` as BREF (Booking Reference)<br>◦ `typeCode` as CONT (Container ID) |

- You must specify the values for the following fields from the shipper system. The below table contains the fields from Provider API for Invoicing and the corresponding reference field from the Consumer API for Freight Subcontracting:

| Field Name in Provider API for Invoicing | Corresponding Field Name in Consumer API for Freight Subcontracting |
| --- | --- |
| purchasingParty | orderingParty > sourceSystemBusinessPartnerID |
| billFromParty | carrier > sourceSystemBusinessPartnerID |
| senderSystemId | originatorId |
| items > Item > transportationItemID | items > Item > id |
| items > item > transportationStageID | transportationStages > TransportationStageItem > senderSystemStageID |
| items > item > pricingElements > logisticalContext | items > Item > id / transportationStages > TransportationStageItem > senderSystemStageID |

- `taxCode` – As the tax codes are country-specific, there is no documentation on valid values for tax codes in SAP Business Network for Logistics. You can manage the tax codes offline with your ordering party.

## Possible Errors

The following are the possible errors that you may get:

- Calculation level of invoice does not match the calculation level of freight document
- No calculation basis present in the freight document
- Currency values are mismatching EUR* USD*
- Charge code TOLLS* is not mapped in the Assign Charge Codes app in the shipper's system
- Business partner LBN_SHIP* does not exist in freight document
- Carrier invoice with this ID exists already
- Final amount cannot be zero for charge lines

- Assign Charge Codes app contains duplicate charge codes TOLLS*
- No charge items are found with the corresponding logistical reference ID and freight document ID
- Rate amount and final amount must be the same for unplanned charge
- Rate amount currency and final amount currency must be the same for unplanned charge
- Quantity will be ignored for an unplanned charge
- Percentage rate is not allowed for an unplanned charge
- Invoice amount cannot be less than zero
- Final amount cannot be zero for charge lines
- Tax amount cannot be negative
- Sum of all the pricing elements and item taxes is not equal to the item gross amount
- Sum of all the tax amount not matching with total tax amount
- The final amount in the invoice and the freight order must be the same for ADJ Charge
- Charge type ADJ cannot be an unplanned charge

* Example values

# Provider API for Invoice Cancellation

As a carrier, you can use this provider API to cancel an invoice when integrating SAP Business Network for Logistics with other systems.

## Description

You can use this inbound API to cancel an invoice that you have created using the Provider API for Invoicing.

When you cancel an invoice, the system creates a credit memo.

## Connection Information

| Technical Name | Field |
|---|---|
| Company Type | Carrier |
| Version | 1.0.0 |
| Format | JSON |
| Prerequisites | Ensure that the invoice document exists in your SAP Business Network for Logistics.Note that this invoice must not be cancelled already and must not have the status as **In Error** or **Rejected**. |
| Predecessors | Recommended: `Provider API for Invoicing` |

You can download the YAML file from the **Provider API for Invoice Cancellation** in SAP API Hub.

## Data Dictionary

In the SAP API Hub, in the **Provider API for Invoicing**, you can view the field details in the **Model View**. In the **API Reference** section, you can view the example values.

In the downloaded YAML file you can view the following details of the fields in the API interface:

- Field name
- Type
- Description
- Max length
- Required or not
- Example

## Additional Information

- Post submission of the API message, login to SAP Business Network for Logistics and launch the **Manage Integration Logs** app. If you see a message with status code 200, then the submission may not be successful because there might be data errors or business validation errors that prevented the creation of invoice in SAP Business Network for Logistics.
- One of the mandatory fields is `lbnInvoiceId`. This is the invoice ID of the freight document in SAP Business Network for Logistics
- If the `creditMemoId` is not sent then the system generates the ID.
- You must specify the values for the following fields from the shipper system. The below table contains the fields from **Provider API for Invoice Cancellation** and the corresponding reference field from the **Provider API for Invoicing**:

| Field Name in Provider API for Invoice Cancellation | Corresponding Field Name in Provider API for Invoicing |
|---|---|
| lbnInvoiceId | carrierInvoiceID |
| orderingPartyLbnId | orderingPartyLbnId |
| carrierLbnId | carrierLbnId |
| senderSystemId | senderSystemId |

## Possible Errors

The following are the possible errors that you may get:

- Invoice does not exist
- Invoice has already been cancelled
- Invoice has already been rejected
- Invoice has the status as API Invoice Error
- Credit memo ID exists already. Enter a different ID

# Planning in External System

The ordering party creates transportation requirements from a sales order, purchase order, or delivery documents in their SAP Transportation Management system. The ordering party can send these transportation requirements to you and request you to perform planning activities in your system. Once you create a planned freight document in your system, you can send the document containing the transportation requirements back to the ordering party.

The ordering party sends the transportation requirements from SAP Transportation Management system to SAP Business Network for Logistics system using the SOAP API `TransportationOrderGenericRequest_Out`.

SAP Business Network for Logistics system sends the transportation requirements to your external system using the **Consumer API to Request for Planning in External System**. You can perform the planning activities in your external system.

You can use the **Provider API to Create Planned Freight Document from External System** to create a planned freight document in the SAP Business Network for Logistics system. SAP Business Network for Logistics uses the SOAP API `TransportationOrderGenericRequest_In` to send this planned freight document to the ordering party's SAP Transportation Management system.

# Consumer API to Request for Planning in External System

As a carrier, you can use this consumer API to receive the transportation requirements sent by the ordering party when the ordering party requests you to perform planning activities in your external system.

You can also use this API to send a request for cancelation of a freight document to the external system.

## Description

The ordering party can send the transportation requirements to you and request you to perform the planning activities in your external system. You can send the planned freight document containing the transportation requirements back to the ordering party.

## Connection Information

| Technical Name | Field |
|---|---|
| Company Type | Carrier |
| Version | 1.0.0 |
| Format | JSON |
| Prerequisites | Ensure that your ordering party has maintained the necessary connections in the Configure Partner Connections app. The ordering party must map the connection details with the business partners and establish a B2B connection. |
| Base Document Type | Type of freight document supported:<br>• 1122 – Transportation Order |

You can download the YAML file from the Consumer API to Request for Planning in External System in SAP API Hub.